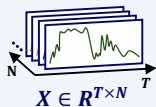


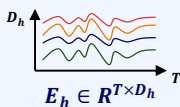
Input Interface

Multi-modal Inputs

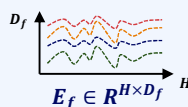
Historical Target Status



Historical Exogenous (ExG_h)



Future Exogenous (ExG_f)



Graph Metadata (M_G)

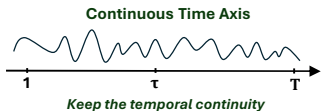


I. Tokenization

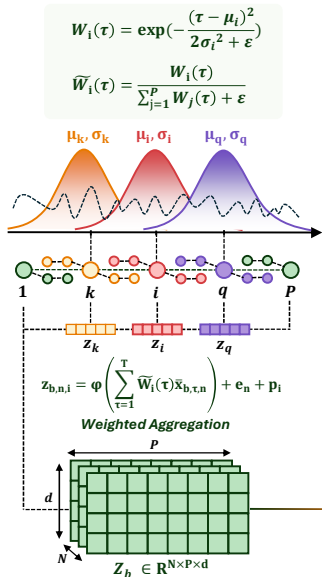
AGSP Continuous Patching

AGSP Module

Adaptive Gaussian-soft-windowed Patching



Soft Overlapping Gaussian Windows

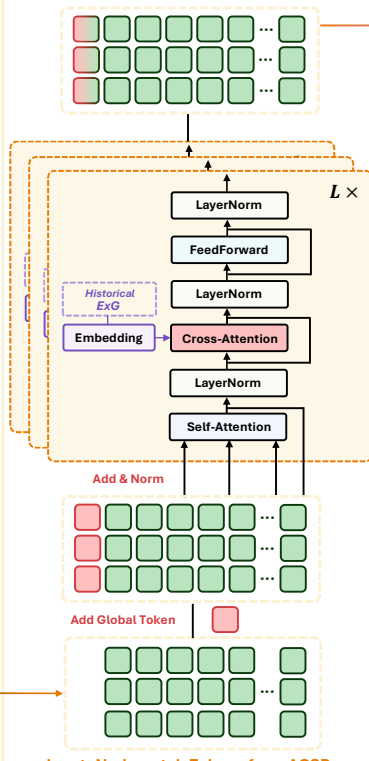


II. Temporal Encoding

Transformer Backbone

Temporal Transformer

Multi-head Self-Attention with Global Token

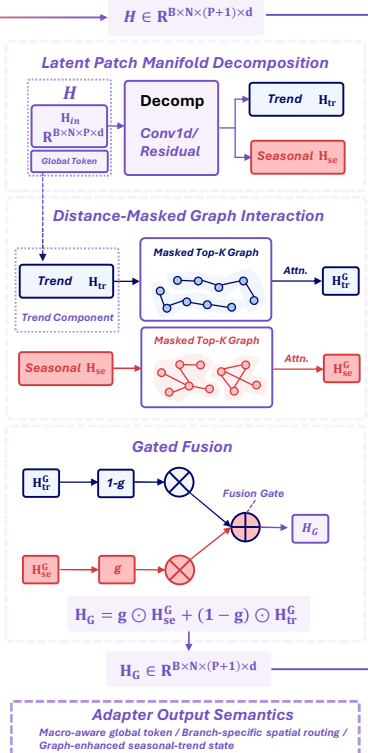


III. Graph Reasoning

R-STMF Adapter

R-STMF Module

Relational Spatio-Temporal Manifold Fusion

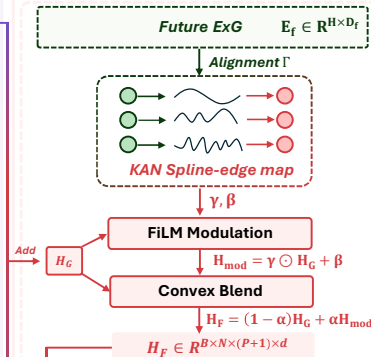


IV. ExG-Conditioned Decoding

K-AMC + PI-DCH

K-AMC Module

KAN-based Exogenous Modulation



PI-DCH Module

Physics-Informed Dynamic Capping

